THOMPSON RIVERS UNIVERSITY

COMP3540 – HUMAN COMPUTER INTERACTION AND DESIGN

TEAM PROJECT PHASE #0: Project Proposal

Team Members: Arshdeep Ghotra (T00684423), Parneet Dhiman (T00705651), Nand Bhasin (T00682801)

Team Introduction

Parneet Dhiman: I am a 3rd year student in Bachelor of Computing Science. I have contributed my role in providing the PROPOSAL. It contains the information about the solution that our website is providing to the users, the technical feasibility, and the project plan that we are looking up to complete our target.

Arshdeep Ghotra: I will primarily work on programming and development. My role involves turning the design concepts into a functioning app, making sure everything runs smoothly for users. I work on efficient task completion, ensuring that the features are seamless. My goal is to bring the technical aspects together to create a responsive and user-friendly FitBuddy experience.

Nand Bhasin: As a team member in the FitBuddy project, my role focuses on UI/UX design, contributing to the creation of a visually appealing and user-friendly interface. Leveraging my intuition in design principles, I aim to enhance the simplicity, intuitiveness, and engaging aspects of FitBuddy. Collaborating with the team, I bring creative solutions in the programming and testing of our website.

BACKGROUND

MOTIVATION

- **WHY IS THIS PROJECT WORTH DOING?**

The fit buddy fitness tracker app project is a timely and essential endeavor, particularly in today's health-conscious society. It addresses a growing trend of health and fitness awareness and meets the increasing demand for technology integration in personal management. Several key reasons are:

1) **HEALTH AND FITNESS TREND:** There's a global shift towards healthier lifestyles, and "fit buddy" taps into this trend by offering a tech-based solution for health and fitness tracking. This aligns with the growing public interest in maintaining health and fitness.
2) **HCI DESIGN AND USER ENGAGEMENT:** The application of human-computer interaction principles in "Fit Buddy" ensures it's not just functional but also user- friendly. HCI design allows for a personalized and adaptive user experience, making the app accessible and engaging for a wide range of users, from fitness enthusiasts to beginners.
3) **PROBLEM SOLVING:** "Fit Buddy" addresses the sedentary lifestyles by encouraging physical activity through interactive technology. It also offers a data-driven approach to fitness, filling the gap in the market for reliable and easy-to-use fitness tracking tools.
4) **WIDER IMPACT:** Beyond individual users, "fit buddy" is a valuable tool for healthcare professionals, providing them with data-driven insights for better patient management. Additionally, the app's

design and user interaction data contribute to HCI research, aiding in the development of future health and fitness technologies.

- **WHAT PROBLEM DOES IT SOLVE?**

1) **PERSONALIZATION IN FITNESS PROGRAMS:** one of the primary issues “fit buddy” solves is the lack of personalization in fitness routines. Unlike generic workout apps, “fit buddy” offers individualized workout plans that are tailored to each user’s unique fitness level, goals and preferences. This personalized approach ensures that users receive a fitness program that is effective and engaging for them especially. Additionally, the app adapts to changes in the user’s fitness level and preferences over time, maintaining the relevance and effectiveness of the workout plans.
2) **MOTIVATION AND ENGAGEMENT IN FITNESS:** To combat the common issue of declining motivation in fitness routines, "Fit Buddy" incorporates elements of gamification and reward systems. These features keep users engaged and motivated in their fitness journey. The app also includes social features, such as community challenges and the ability to connect with friends, fostering a sense of community and accountability, which are crucial for sustained engagement in fitness activities.

3) **ACCESSIBLITY TO FITNESS GUIDANCE:** "Fit Buddy" is particularly beneficial for beginners in fitness, offering guided workouts and instructional content that make it easier for them to start and maintain a fitness routine. The app is also designed with accessibility in mind, ensuring that users of all abilities can easily navigate and utilize its features.

- **WHO DOES IT HELP?**

The “fit buddy” fitness tracker app is designed to assist a diverse range of users, each with their unique health and fitness needs:

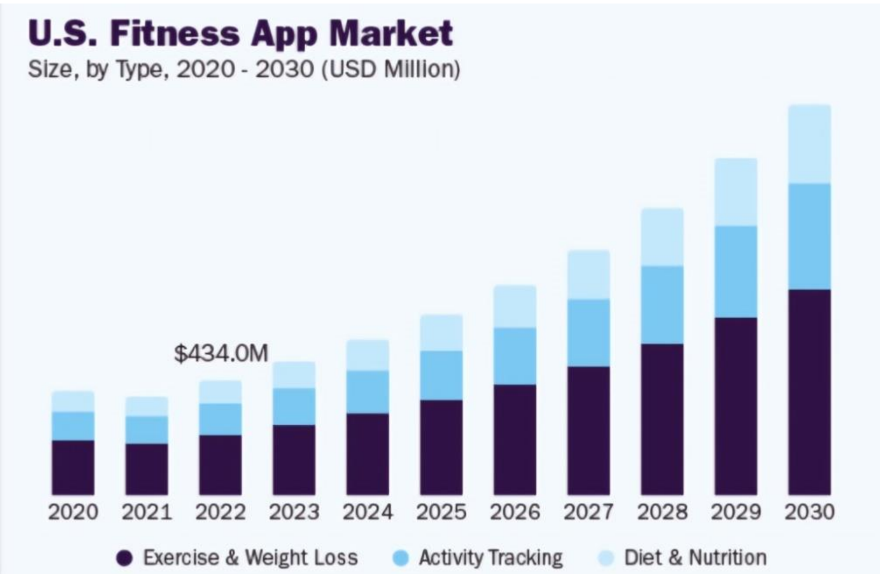
1)**FITNESS ENTHUSIASTS:** For those deeply invested in their fitness journey, "Fit Buddy" provides advanced tracking and analysis features. It offers detailed insights into workouts and health metrics, essential for optimizing performance. Additionally, the app allows enthusiasts to set and adjust fitness goals, catering to their evolving fitness aspirations.

2) **BEGINNERS IN FITNESS:** The app is particularly beneficial for beginners. It reveals the fitness world to guided workouts and instructional content, making the initiation into fitness less intimidating. "Fit Buddy" also features programs that progressively increase in difficulty, accommodating the evolving capabilities of new users.

3) INDIVIDUALS WITH SPECIFIC HEALTH GOALS: For users focused on weight management or dealing with chronic conditions like diabetes, "Fit Buddy" offers tailored diet and exercise plans. These personalized features ensure that users with specific health objectives receive appropriate guidance and support.

RESEARCH

1. **MARKET RESEARCH:** the fitness app market is a dynamic and rapidly =growing sector, as highlighted in the initial summary. In-depth market research indicates that the demand for fitness apps is driven by various factors, including the increasing prevalence of mobile devices, rising health consciousness and the desire for convenience in tracking fitness activities. As more people prioritize health and fitness, there is a clear need for technology-driven solutions to assist them in achieving their goals.



2. **ENVIRONMENTAL ANALYSIS:** Analysis reveals that societal trends, such as the increasing urban population, sedentary lifestyles, and concerns about physical and mental health, are contributing to the

growth of this market. The COVID-19 pandemic has also accelerated the adoption of fitness apps, as people seek ways to stay active and healthy while adhering to social distancing.

3. **Literature review:** Numerous studies and reports support the findings of the market research. Research from organizations like Statista and Grand View Research consistently project substantial growth in the fitness app market. Additionally, academic studies have shown that fitness apps can be effective in prompting physical activity and improving overall well-being.

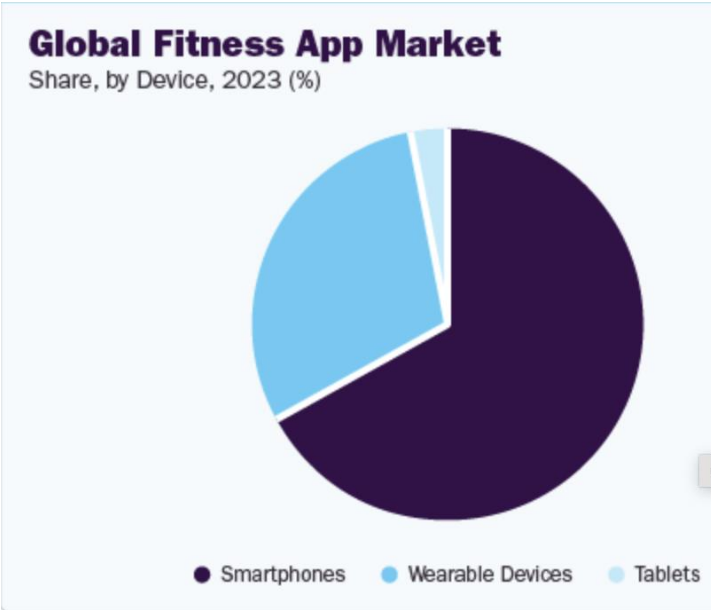
EXISTING PRODUCTS:

Several fitness tracker apps and wearable devices are already prominent in the market. Notable examples include Fitbit, Apple Watch with the associated with fitness+ app, Garmin, and Samsung Health. These products utilize a combination of sensors, including accelerometers, heart rate monitors and GPS technology, to track various fitness metrics.

The need for fitness tracker apps persists due to their ability to provide personalized fitness tracking, offer guided workouts. Users appreciate the convenience of having a personal trainer or health advisor on their devices, especially during the ongoing trend of remote and home-based fitness routines.

TECHNOLOOGY:

Fitness tracker apps leverage a range of technologies, including mobile app development, wearable device connectivity, cloud computing and data analytics. These apps rely on sensors within smartphones and wearable devices to collect data on steps taken, heart rate, sleep patterns and more. Advanced apps may also incorporate machine learning and AI algorithms to provide tailored fitness recommendations and predict user preferences.



Smartphones: The dark purple segment takes up most of the chart. This looks to be approximately 70-75%.

Wearable Devices: The light blue segment is significantly smaller than the smartphone segment, around 20-25%.

Tablets: The smallest segment, in white, appears to be about 5-10%.

TARGET USERS:

1) MELLENNIALS (AGE 25-40):

- Fitness enthusiasts: Millennials who are dedicated to their fitness goals and appreciate data-driven insights to track their progress.
- Busy Professionals: this age group often has demanding work schedules and values time-efficient workouts that fit into their busy lives.
- Health- Conscious: Many millennials are health-conscious, seeking a holistic approach to fitness that includes both physical and mental well-being.

2) GENERATION X (AGE 41-56):

- Fitness Enthusiasts: Some individuals in this age group maintain active lifestyles and are dedicated to their fitness routines.
- Beginners and Novices: Those looking to adopt healthier habits later in life may appreciate user-friendly guidance and ease of use.
- Health-Conscious: Gen Xers increasingly focus on overall health and wellness, making them interested in comprehensive fitness tracking.

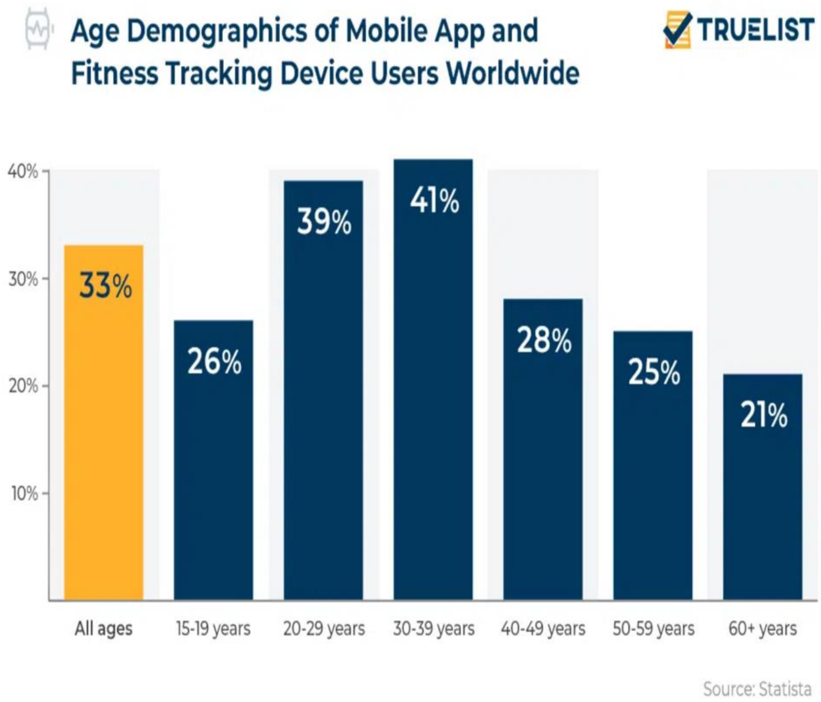
3)BABY BOOMERS (AGE 57-76):

- Seniors: Baby boomers may value fitness apps that cater to their specific needs, including gentle exercises and adaptability to aging bodies.
- Health-Conscious: This age group places a premium on maintaining good health as they age, making features like stress management and sleep tracking relevant.

This age group places a premium on maintaining good health as they age, making features like stress management and sleep tracking relevant.

4) Generation Z (Age 18-24):

- Fitness Enthusiasts: Some younger users are highly dedicated to fitness and may seek advanced tracking and workout plans.
- Beginners and Novices: Others in this group may be just starting their fitness journey and appreciate a user-friendly interface and guidance.


Age Demographics of Mobile App and Fitness Tracking Device Users Worldwide
TRUELIST
40%
30%
20%
10%
33%
26%
39%
41%
28%
25%
21%
All ages
15-19 years
20-29 years
30-39 years
40-49 years
50-59 years
60+ years
Source: Statista

PROPOSAL

SUPPORT (SERVICE) OR SOLUTION

Following are the solutions and services to the problems which our website provides to the users in their day-to-day life regarding health issues:

- Issue: A sedentary lifestyle and insufficient physical exercise.
- Fitness programs, exercise schedules, and dietary guidance may all be found on our website. They can inspire users to lead better lifestyles by incorporating features like goal setting and activity monitoring.

- Issue: An unhealthy and imbalanced diet.
- Offer users with tools or features that enable them to develop customized diet programs that take into account their unique health objectives, food preferences, and any underlying medical issues. This can be especially helpful for those who are trying to follow a certain diet plan, control their weight, or treat nutritional deficiencies.

- Issue: More concerns about mental health.
- Providing audio files, video lessons, or guided meditation sessions might assist users—especially newcomers—in learning and practicing meditation efficiently. These tools may be customized to meet a variety of purposes, including relaxation, mindfulness, and stress reduction.

- Issue: Lack of communities that offer assistance to those dealing with health issues.
- Setting up online communities and forums where people can ask for help, discuss their experiences, and offer emotional support. Users will be encouraged to participate in challenges, compete with their friends and win rewards. These actions and networks can lessen feelings of loneliness, gives motivation and provide a sense of community.

TECHNICAL FEASIBILITY

Creating a health-related website that offers holistic approach to fitness and well-being involves several steps. Below is a comprehensive guide to help you through the process:

- Analyzing the market: Analyze existing health and wellness websites to understand trends and user preferences.
- Defining the purpose and target audience: Clearly outlining the purpose of our website (e.g., promoting overall health and well-being) and identifying the target audience (e.g., beginners, fitness enthusiasts, individuals interested in meditation).
- Content Planning: Creating a content plan that includes workout routines, diet plans, guided meditation sessions, and community engagement content and developing a calendar or a tracker to ensure regular updates.
- Deciding the tools and technology to use: Decide on the technology stack for our website and ensure compatibility with browsers and mobile devices.
- Designing a user-friendly platform: Creating a user-friendly and responsive design that works well on various devices and focusing on intuitive navigation to enhance the user experience.
- Security and privacy: Clearly communicating the privacy policy and implementing security measures to protect user data.
- Prototyping and testing: Creating a prototype and testing with a pool of target users to get a feedback about the positive as well as the lacking points of the website to amend the changes before the final launch.

Language/libraries/frameworks that will be used to incorporate the plan:

We will potentially use the following

- Front end: HTML, CSS, JavaScript. Framework: React.Js
- Back end: Python. Framework: Django
- Database: SQL
- For styling: Bootstrap, Tailwind CSS
- Platform for hosting and deployment.

PROJECT PLAN

We are planning to execute our project plan, biweekly:

WEEK 1-2: doing the programming and creating the interface that the user will be interacting with (Front-end).

WEEK 3-4: creating the database and networking of the components of website for the developer's use (Back-end).

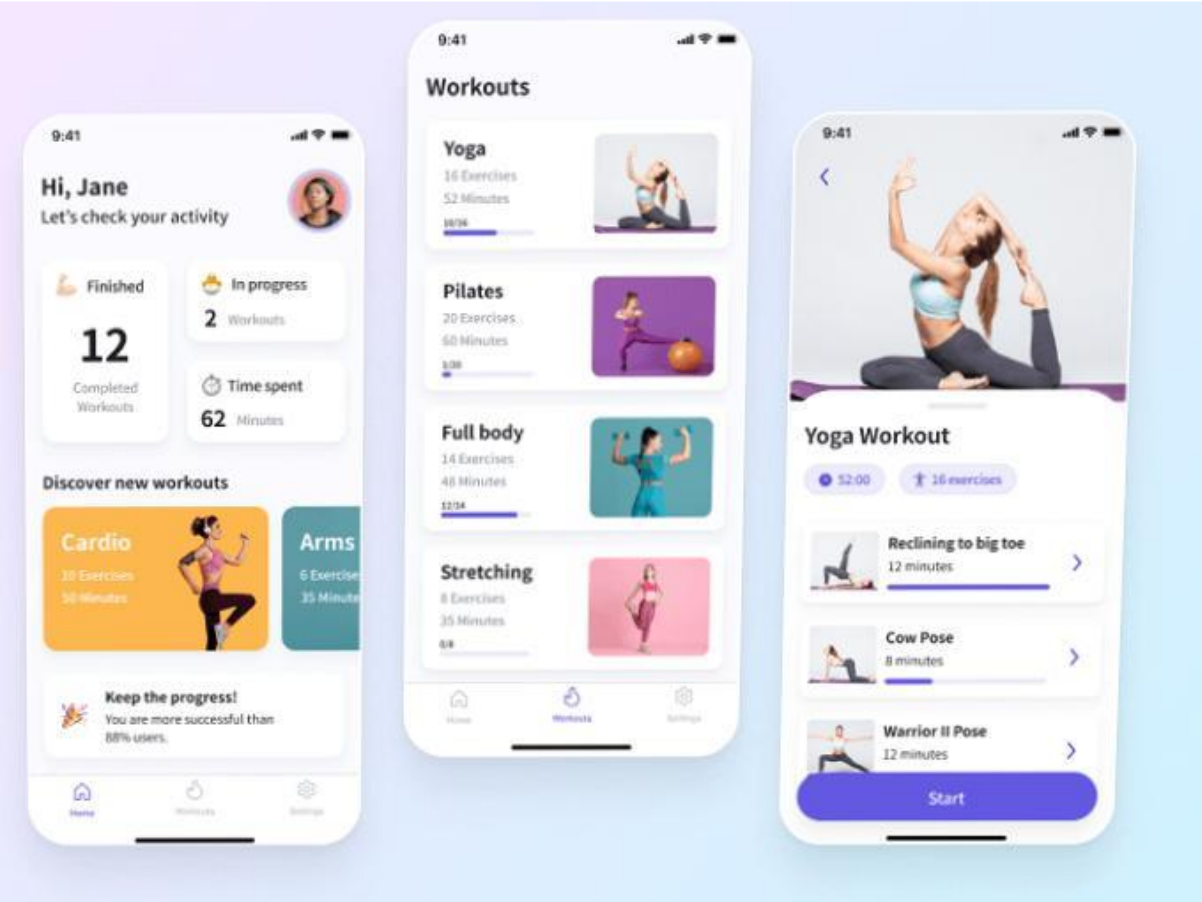
WEEK 5-6: Testing the product, doing the documentation, and launching our final WEBSITE.

GOALS

Usability Goals

Simplicity and Intuitiveness: Ensuring the users can effortlessly navigate FitBuddy is essential. Our goal is to create an interface that caters to users with varying technical abilities for efficient app usage. To achieve thisgoal, our strategy revolves around employing a minimalistic design coupled with clear and concise navigation. By minimizing complexity in visual elements and navigation pathways, we aim to enhance the app's intuitiveness. We prioritize to create an interface for a seamless and user-friendly experience.

An example of minimalistic UI:



Efficient Task Completion: potential users interact with FitBuddy with a clear goal in mind, which is, accomplishing specific fitness tasks. Therefore, it's crucial that the website facilitates swift and straightforward task completion. To achieve this, our strategy involves implementing a streamlined workflow for key tasks (and all around the user interface) such as starting a workout, logging nutrition, engagin with other users and reviewing progress. We aim to minimize the number of steps required to achieve common objectives. This strategy not only line-up with the user's intent but also enhances the overall efficiency of FitBuddy, ensuring that users can seamlessly and quickly accomplish their fitness-related tasks.

Consistency Across Platforms: Maintaining consistency in FitBuddy's usability across various devices, particularly mobile and web, is crucial to provide users with a seamless experience when transitioning between platforms. The relevance lies in ensuring that regardless of the platform, users experience a familiar interface. Our strategy involves following design guidelines specific to different platforms. This approach ensures a consistent visual language and consistent interaction patterns across diverse devices.

User Feedback Integration: we acknowledge the value that user feedback holds for the continuous improvement of our app. user insights play a crucial role in improvement of our app. We will actively seek and integrate user suggestions and concerns to ensure FitBuddy aligns with their needs and expectations. Our strategy involves the integration of feedback mechanisms directly within the app, such as surveys and a dedicated feedback section. By establishing a feedback loop, we encourageusers to actively contribute to the refinement and enhancement of FitBuddy.

UX Goals

Engaging and Motivational Design: FitBuddy is all about creating an interface that goes beyond just being user-friendly – it's about keeping users hooked and motivated throughout their entire fitness journey. our strategy involves seamlessly integrating gamification elements into the app. Badges, level progression, and personalized achievements (some examples) become not just features but milestones in the user's fitness journey. we aim to incorporate vibrant visuals and use encouraging language to create a positive user experience.

Personalization for User Empowerment: We recognize that fitness is a deeply personal journey, and FitBuddy is committed to being more than a one-size-fits-all solution. Our strategy is to allow users to shape their fitness experience. From personalized workout recommendations to customizable dashboards and adaptive notifications, these features are designed to encourage users tocustomize their experience according to their unique preferences.

Community Building Through Social Integration: Fitness is not just a solitary journey, but often a shared experience. We understand that a sense of community is a powerful motivator, encouraging users to not only celebrate their own achievements but also to support one another. To bring this vision to life, our strategy revolves around seamlessly incorporating social features into the app. Features such as achievement sharing, friend challenges, and community forums help us create virtual spaces where FitBuddy users can come together. Through these features, we not only encourage the sharing of accomplishments but also facilitate positive interactions and the exchange of knowledge and insights.

An example of gamification and social integration.